

Lecture 17

Monday, April 13, 2026 7:22 PM

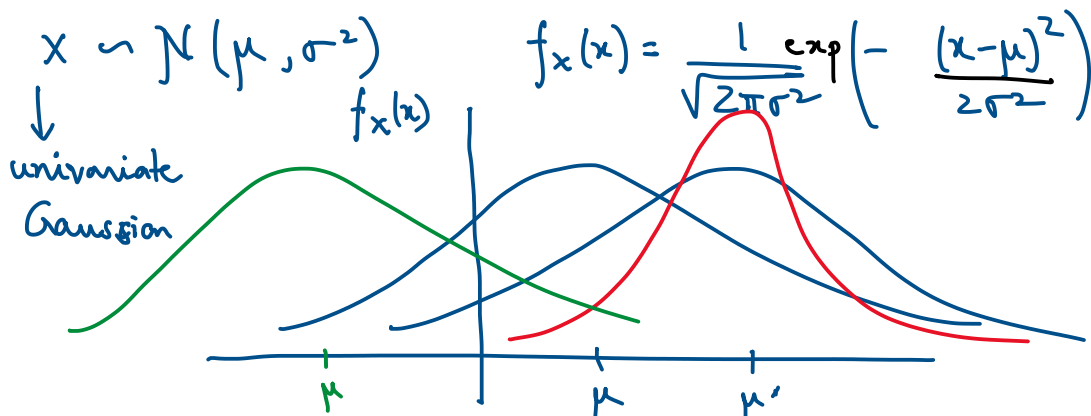
Recap: Logistic Regression (binary classification)

- ① Training data $(x^{(i)}, y^{(i)})$ $1 \leq i \leq N$, $x^{(i)} \in \mathbb{R}^d$
 $y^{(i)} \in \{0, 1\}$
- ② $h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}} = p(y|x; \theta)$
- ③ $\hat{y} = \begin{cases} 1 & \text{if } \theta^T x \geq 0 \\ 0 & \text{o/w} \end{cases}$
- ④ Cost function: $J(\theta) = -\frac{1}{N} \sum_{i=1}^N y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \leftarrow$
↓
cross entropy loss
- ⑤ $\theta^* = \operatorname{argmin}_{\theta} J(\theta)$: found using G.D.

Questions ?

Today: Gaussian Discriminant Analysis

Prerequisite: The Gaussian Distribution



Multivariate Gaussian (d-dimensional)

$$x \sim \mathcal{N}(\mu, \Sigma)$$

↓ ↑
mean covariance
vector

$\mu \in \mathbb{R}^d$

$\Sigma \in \mathbb{R}^{d \times d}$ ($d \times d$ dimensional matrix)

$$f_x(x) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (x-\mu)^T \Sigma^{-1} (x-\mu)\right)$$

$$f_x(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}\right) \quad \text{Univariate Gaussian}$$

$d=2$ $\Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{pmatrix}$ $\sigma_i^2 = \text{var}(x_i)$
 $\sigma_{12} = \text{cov}(x_1, x_2)$

$$\sigma_{12} = \mathbb{E}[(x_1 - \mu_1)(x_2 - \mu_2)] \quad \leftarrow$$

$$\sigma_1^2 = \mathbb{E}[(x_1 - \mu_1)^2]$$

$$\sigma_2^2 = \mathbb{E}[(x_2 - \mu_2)^2]$$

$$\sigma_{12} \rightarrow \mathbb{E}[(x_1 - \mu_1)(x_2 - \mu_2)] = \sigma_{21}$$

$\sigma_{12} > 0 \Rightarrow$ at higher values of x_1 ,
 higher values of x_2 are more
 likely than lower values of x_2

$\sigma_{12} < 0 \Rightarrow x_1 \text{ higher} \Rightarrow x_2 \text{ lower}$

Prerequisite: The Bayes' Rule

x - feature $\in \mathbb{R}^d$
 y - $\in \{0, 1\}$

$P(y|x)$
 $\uparrow$ class
 $\uparrow$ feature
 $\uparrow$ Logistic regression

$P(x|y)$
 $\downarrow$ feature
 $\rightarrow$ class
 $\uparrow$ GDA

$$P(y|x) = \frac{P(x|y) P(y)}{P(x)} \quad \} \text{ Bayes' Rule}$$

GDA: Gaussian Discriminant Analysis

Q. Given these features, which class does the sample belong to?

GDA: "What a typical sample from each class look like?"

$$P(x|y=1) \text{ \& } P(x|y=0)$$

Modeling Assumptions under GDA (binary)

① Class $y \sim \text{Ber}(\phi)$ (Bernoulli)

$$\Rightarrow P(y=1) = \phi$$

$$P(y=0) = 1-\phi$$

② Within each class, features are multivariate Gaussian

$$\left. \begin{aligned} x|y=0 &= \mathcal{N}(\mu_0, \Sigma) \\ x|y=1 &= \mathcal{N}(\mu_1, \Sigma) \end{aligned} \right\} P(x|y)$$

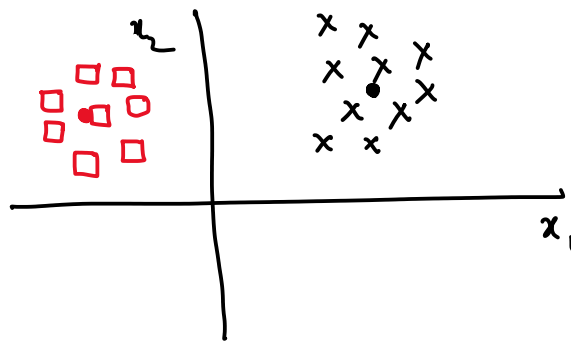
Note that Σ is the same for both classes

Parameters: $\phi, \mu_0, \mu_1, \Sigma \rightarrow O(d^2)$

$$x^{(i)} \in \mathbb{R}^d$$

$$y^{(i)} \in \{0, 1\}$$

$$\mu_0 = \begin{bmatrix} E[x_1] \\ E[x_2] \end{bmatrix}$$



x class 0
□ class 1

Open questions:

① How to find the "best" $\mu_0, \mu_1, \phi, \Sigma$?

② How to determine $\hat{y}$ for new sample?

Ans: To classify a new point

$$\hat{y} = \underset{\hat{y}=1}{\operatorname{argmax}} P(y|x)$$

new point

$$\equiv \mathbb{P}(y=1|x) \underset{\hat{y}=0}{\geq} \mathbb{P}(y=0|x) \quad - (1)$$

(recall: we have modeled $\mathbb{P}(x|y)$)

$$\text{LHS} = \frac{\mathbb{P}(x|y=1) \mathbb{P}(y=1)}{\mathbb{P}(x)}$$

$$\text{RHS} = \frac{\mathbb{P}(x|y=0) \mathbb{P}(y=0)}{\mathbb{P}(x)}$$

$$(1) \Rightarrow \underbrace{\mathbb{P}(x|y=1)}_{\mu_1, \Sigma} \underbrace{\mathbb{P}(y=1)}_{\phi} \underset{\hat{y}=0}{\geq} \underbrace{\mathbb{P}(x|y=0)}_{\mu_0, \Sigma} \underbrace{\mathbb{P}(y=0)}_{1-\phi} \quad - (2)$$

$$\mathbb{P}(x) = \mathbb{P}(x|y=1) \mathbb{P}(y=1) + \mathbb{P}(x|y=0) \mathbb{P}(y=0)$$

(not necessary for classification)

Ans to (1)

Write the likelihood function, $\arg \max$

$$\mathcal{L}(\phi, \mu_0, \mu_1, \Sigma) = \prod_{i=1}^N \mathbb{P}(x^{(i)}, y^{(i)})$$

$$= \prod_{i=1}^N \underbrace{\mathbb{P}(x^{(i)} | y^{(i)})}_{\text{multivariate gaussian}} \underbrace{\mathbb{P}(y^{(i)})}_{\text{Bernoulli}}$$

$$(\phi^*, \mu_0^*, \mu_1^*, \Sigma^*) = \arg \max \mathcal{L}(\phi, \mu_0, \mu_1, \Sigma).$$

Solution (omit derivation)

① ϕ^* = fraction of ^{training} data samples that are class 1.

② μ_0^* = sample mean of the class 0 samples in the training data

③ μ_1^* = sample mean of the class 1 samples in the training data

$$\textcircled{4} \Sigma^* = \frac{1}{N} \sum_{i=1}^N (x^{(i)} - \mu_{y(i)}) (x^{(i)} - \mu_{y(i)})^T$$

sample covariance across both classes

Using GDA, the decision boundary is linear.

$$\underbrace{(\mu_1 - \mu_0)^T \Sigma^{-1} x}_{\theta^T} + \underbrace{\left(-\frac{1}{2} \mu_1^T \Sigma^{-1} \mu_1 + \frac{1}{2} \mu_0^T \Sigma^{-1} \mu_0 + \log \frac{\phi}{1-\phi} \right)}_{\theta_0} = 0$$

$$\downarrow \quad \downarrow$$

$$\theta^T x + \theta_0 = 0$$

(hyperplane)

Proof outline:

$$\underbrace{P(y=1|x)}_{\text{substitute } f_x(x)} \geq \underbrace{P(y=0|x)}_{\text{substitute } f_x(x)}$$

$$\underbrace{P(x|y=1) P(y=1)}_{\text{substitute } f_x(x)} \geq \underbrace{P(x|y=0) P(y=0)}_{\text{substitute } f_x(x)}$$

